

The Broad Center Superintendent Research Dataset

V2: May 12, 2026

1 Overview

This document contains figures and tables generated using the Superintendent Research Dataset.

The data described here are in the file `data/processed/combined_superintendents.csv`. All figures and tables are produced in `03_tables_figures.R` and can be found in the folder `/scripts`. See the README file for details on data generation.

You are welcome to use these data. Please do so with the following citation:

Stemper, Sam and The Broad Center. Superintendent Research Dataset (v2, 2026-05-12), 2026.

Statistics may differ somewhat from Stemper (2022) due to updates in the dataset.

2 Coverage

The dataset includes records for 23 states. Figure 1 displays years of coverage and data types available for each state. Figures 2 and 3 report, respectively, the number of districts for which we have data in each state-year combination and the number of districts for which we have a balanced panel of data across all available state-years.

Figure 4 displays the ratio of total enrollment in districts for which we have data to total public school enrollment in a) covered state-years, and b) the US as a whole. In the state-years where we have data, coverage is nearly complete. Our data cover a majority of US school enrollment from 2010-2022. For example, as reported in Table 1, in 2014, covered districts account for 71.8% of public school enrollment and 72.8% of school districts in the US.

3 Benchmarking

We benchmark descriptive statistics in our data to published results.

We first consider superintendent turnover. We compute turnover using our superintendent identifiers, which are created by matching superintendents on name within states. Figure 5 reports turnover rates for each year.

Table 2 compares our estimated turnover rates for the years 2020-2022 to those reported by White (2023) at the national level. Though the samples are different (White (2023) uses national data and we use data from the states in our sample), our estimates are similar.

We next consider superintendent gender. We code superintendent gender on the basis of names. Table 3 reports state-by-state comparisons to The Superintendent Lab (2025). An exact comparison is not possible here because The Superintendent Lab (2025) reports results for 2024-25 and our data go through 2023-24. However, overall levels are similar.

4 Known issues

4.1 Turnover rates

Figure 6 reports turnover estimates by state and year. There are some state-year combinations where turnover rates are non-missing but very low. For example, New York in 2017 or Oklahoma 2020 fit this description. This is likely because the data snapshots used to construct superintendent rosters in those state-year cells come from nearby months within consecutive academic years.

4.2 Salaries

Some reported salary values are implausibly low for full-year superintendent positions. We keep these values in the dataset for transparency but note that entries of 0 or very low salaries in some states may not reflect actual full-year compensation.

4.3 Disambiguation

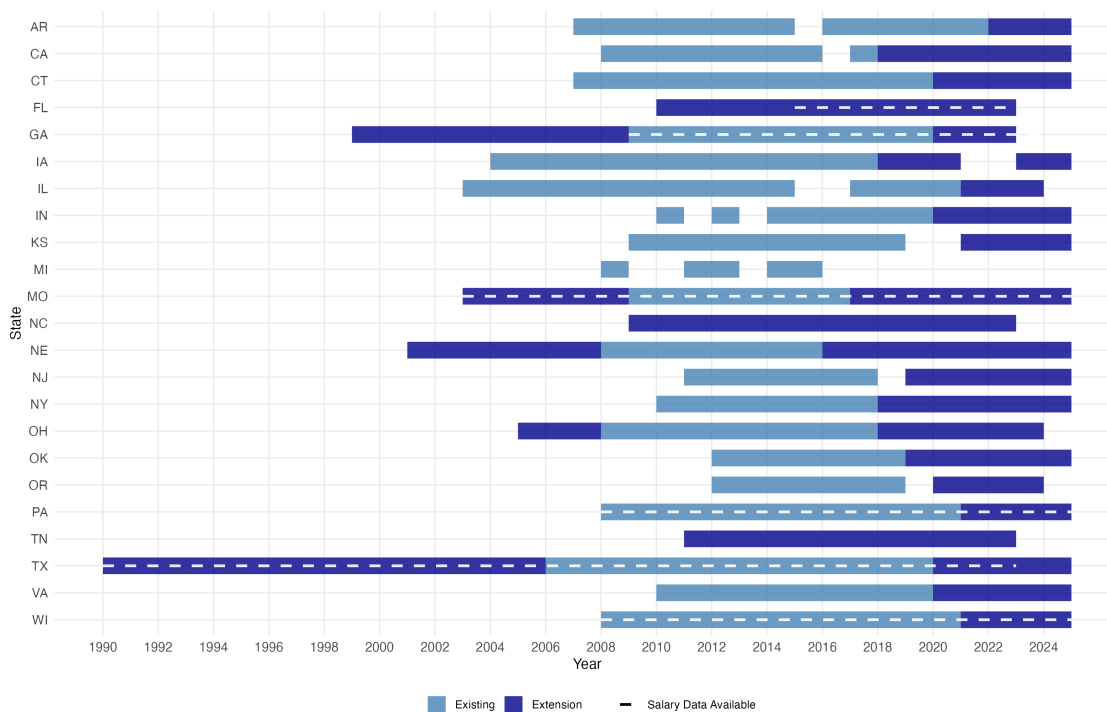
Because the same superintendent may have served in multiple states, our within-state identifiers alone cannot link a person’s records across state lines. To address this, we developed a disambiguation tool (in the `disambiguation/` folder) that determines whether two records sharing the same name—but in different states—refer to the same individual. The tool works by prompting a large language model to search the web for corroborating evidence (news articles, school board minutes, district websites, archived pages) connecting the two tenures. A second “critic” model then independently scores each proposed match on ten criteria, accepting it as confirmed only when the evidence is strong and internally consistent. Out of 1,512 candidate cross-state name pairs examined, 264 were confirmed as the same person.

Once confirmed matches are identified, the script `scripts/04_assign_global_ids.py` merges the within-state identifiers into a single `global_id` for each superintendent. It does this using a union-find algorithm: if superintendent A’s Arkansas record and Missouri record are confirmed as the same person, all of their rows—across both states—are assigned one shared identifier. To assess accuracy, we manually reviewed a random sample of 200 cases by looking

up each superintendent online. The model achieved 99.0% overall accuracy, 97.4% precision (very few false positives), and 100% recall (no confirmed cross-state superintendent was missed).

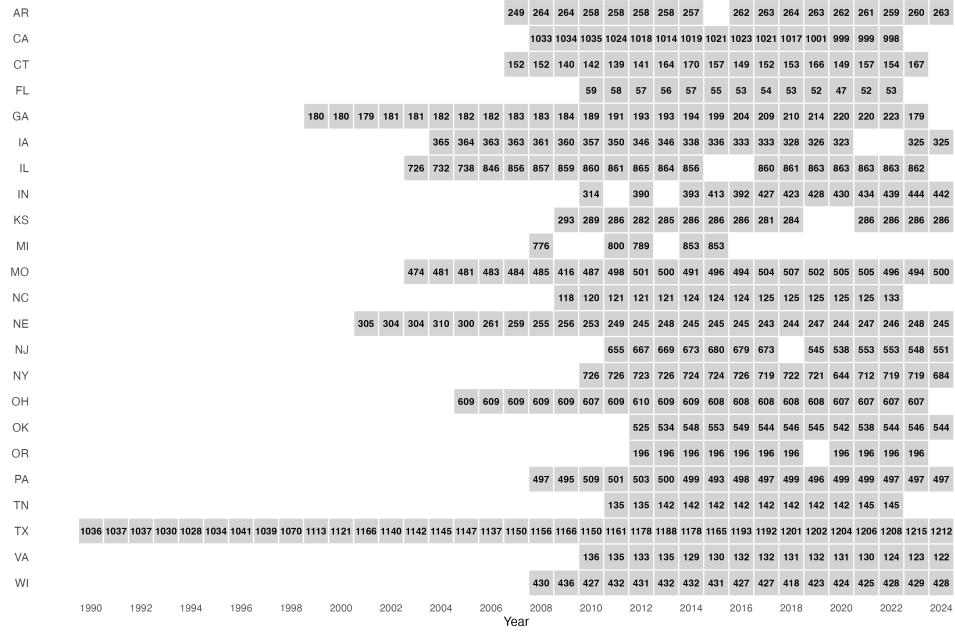
Figures

Figure 1: Data coverage



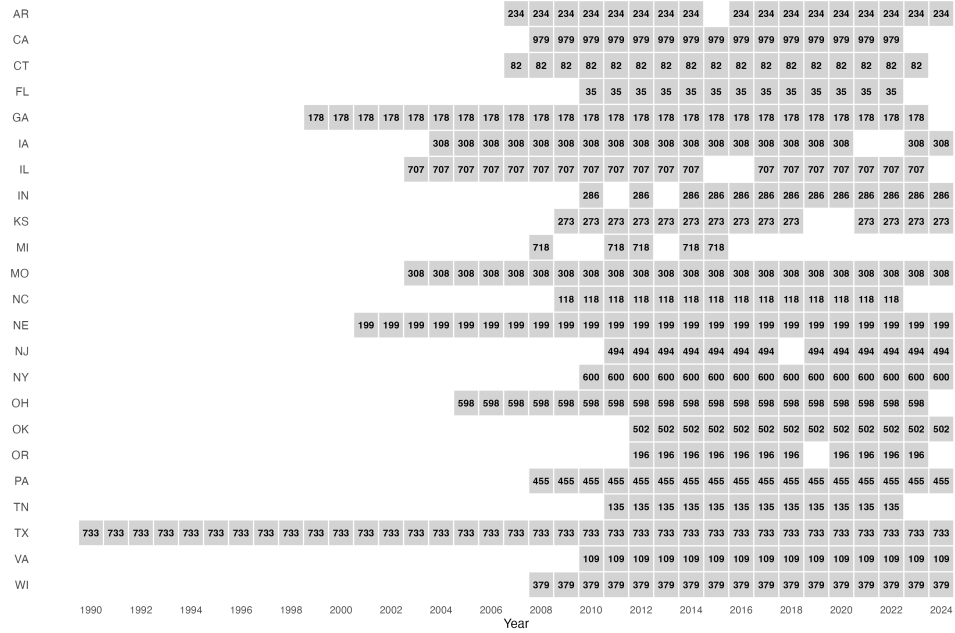
Notes: Year refers to the fall of the academic year. “Existing” data are those used in Stemper (2022).

Figure 2: District counts by state and year



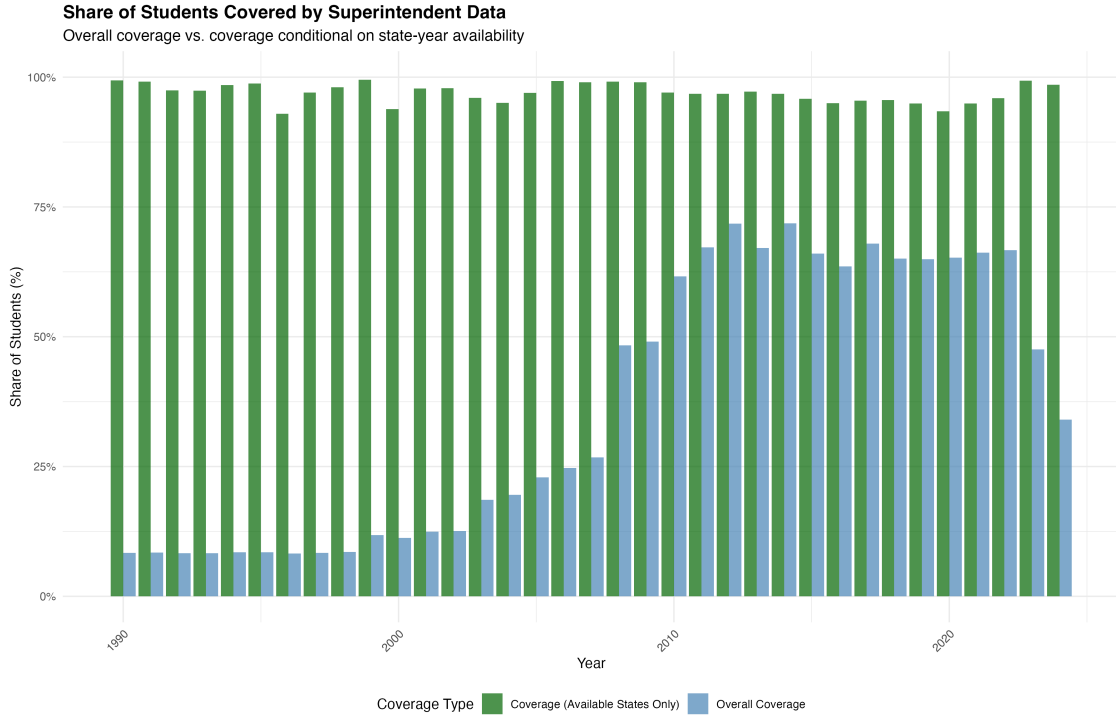
Notes: This figure shows the number of districts for which superintendents data is available.

Figure 3: Balanced district counts by state and year



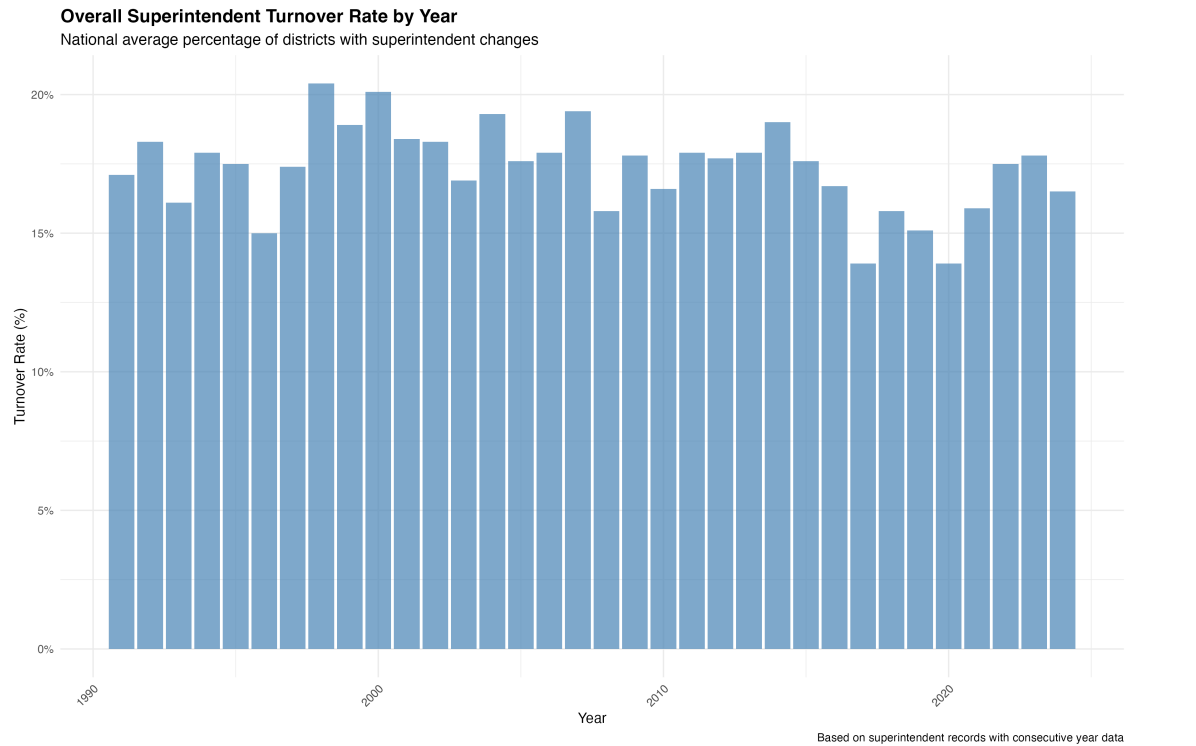
Notes: This figure shows the number of districts for which superintendents data is available in every year for which we have data for that state.

Figure 4: Enrollment coverage by year



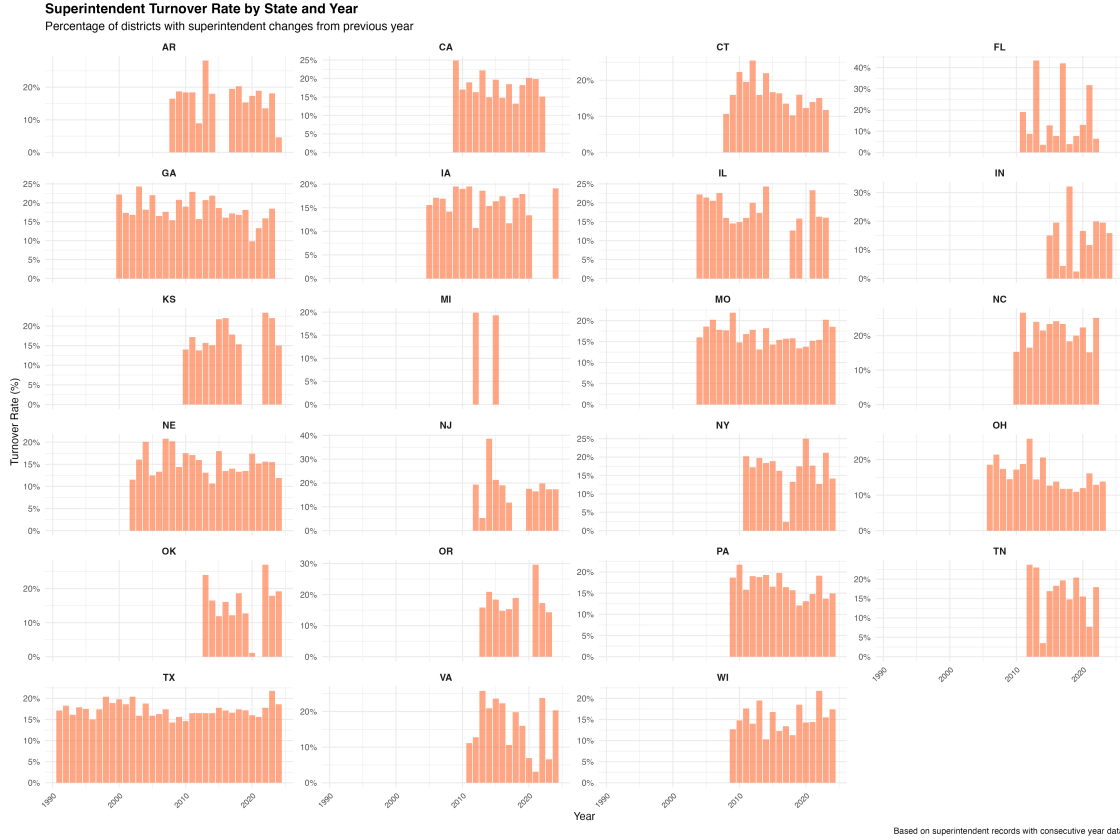
Notes: This figure shows the share of students covered in our data in the country overall (blue bars) and in states for which we have superintendents data in that year (green bars). Coverage is calculated by dividing the student enrollment counts in regular public school districts in our data by total enrollment in regular public school districts. Enrollment data come from the Common Core of Data (CCD). Districts not labeled as “regular public school district[s]” by the CCD are excluded from these calculations.

Figure 5: Superintendent turnover by year



Notes: A district is considered to have a superintendent turnover if the superintendent name in a district changes from one year to the next. We use a string similarity score to allow for minor changes in name spelling or formatting across years. The turnover status for a given district and year is set to missing if the previous year's data is not available for that district.

Figure 6: Superintendent turnover by state and year



Notes: A district is considered to have a superintendent turnover if the superintendent name in a district changes from one year to the next. We use a string similarity score to allow for minor changes in name spelling or formatting across years. The turnover status for a given district and year is set to missing if the previous year's data is not available for that district.

Tables

Table 1: Enrollment and district coverage, 2014

State	Total Enrollment	Enrollment Coverage (%)	Total Districts	District Coverage (%)
AR	479,027	100	237	99.2
CA	6,156,897	100	992	99.2
CT	509,248	97.8	170	94.7
FL	2,745,223	64.3	70	77.1
GA	1,717,805	100	180	100
IA	505,311	100	346	97.7
IL	2,046,476	99.9	862	98.3
IN	1,004,622	100	291	99.7
KS	496,920	100	307	93.2
MI	1,350,667	100	543	100
MO	897,427	93.6	520	91
NC	1,465,031	100	115	100
NE	312,281	100	251	97.6
NJ	1,363,220	100	588	99.3
NY	1,637,426	100	688	100
OH	1,601,836	99.9	615	99
OK	671,715	100	517	99.8
OR	566,951	100	181	98.9
PA	1,589,440	99.9	501	99.4
TN	995,475	99.5	146	96.6
TX	5,003,912	99.2	1,026	98.2
VA	1,265,944	98.4	130	97.7
WI	860,784	99.4	422	98.6
US	47,489,409	71.8	13,080	72.8

Notes: This table shows total enrollment and district counts by state and in our dataset in 2014. As in Figure 2, these calculations only include regular public school districts, as designated by the CCD. This figure is a replication of Stemper, 2022 Table 1 (re-produced as Table 7 in this document).

Table 2: Turnover: comparison to White (2023)

State	2020 (%)	2021 (%)	2022 (%)
AR	17.3	18.9	13.5
CA	20.2	19.9	15.1
CT	12.3	14.0	15.1
FL	13.0	31.8	6.4
GA	9.8	13.3	15.9
IA	13.4	NA	NA
IL	0.0	23.3	16.3
IN	16.6	11.7	19.9
KS	NA	NA	23.4
MO	13.8	15.2	15.4
NC	22.4	15.2	25.2
NE	17.4	15.2	15.6
NJ	17.6	16.5	19.8
NY	25.0	17.7	12.7
OH	12.0	16.1	12.9
OK	1.1	0.0	27.0
OR	NA	29.6	17.3
PA	13.1	14.8	19.1
TN	15.5	7.7	17.9
TX	16.0	15.6	17.8
VA	6.9	3.1	23.8
WI	14.3	14.4	21.8
Total	13.9	15.9	17.5
National	14.2	16.9	17.1

Notes: A district is considered to have a superintendent turnover if the superintendent name in a district changes from one year to the next. We use a string similarity score to allow for minor changes in name spelling or formatting across years. The turnover status for a given district and year is set to missing if the previous year's data is not available for that district. Total turnover rates refer to the total for that year in our data. National turnover rates come from White (2023) Appendix Table S2.

Table 3: Share of male superintendents (%): comparison to The Superintendent Lab (2025)

	The Broad Center (2023-24)	Sup. Lab (2024-25)
AR	69.4	73.8
CT	66.3	62.9
GA	70.1	71.4
IA	85.5	86.9
IL	69.2	71.8
IN	65.8	79.3
KS	74.5	80.5
MO	73.2	67.1
NE	87	85.8
NJ	65.6	65.6
NY	70.7	71
OH	81.3	83.1
OK	73.5	78.7
OR	63.9	67.3
PA	75.2	75
TX	68.4	74.6
VA	62.5	64
WI	72.7	74.8
Total	71.9	74.1

Notes: Gender is predicted based on the superintendent name using the **gender** package in R. The right column is from The Superintendent Lab, 2025.

References

- Stemper, S. (2022). Doing more with less: School management and education production. https://samstemper.com/files/stemper_jmp.pdf
- The Superintendent Lab. (2025). NLSD Data Visualizations [Accessed: 2025-08-08]. <https://thesuperintendentlab.com/communicating-our-work/nlsd-data-visualizations/>
- White, R. S. (2023). Ceilings Made of Glass and Leaving En Masse? Examining Superintendent Gender Gaps and Turnover Over Time Across the United States. *Educational Researcher*, 52(5), 272–285. <https://doi.org/10.3102/0013189X231163139>